

Bishal Roy

Applied AI and Machine Learning | LLM Systems, Model Evaluation, Multimodal Learning

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EDUCATION

B.E. in Artificial Intelligence and Data Science 2022 - 2026 | CGPA 8.99/10

Dr. D. Y. Patil Institute of Technology, Pune

- Final-semester grade 9.55/10; aggregate 8.99/10 across eight semesters.
- **Undergraduate research:** predicting problematic internet use in children from clinical and wearable-sensor data; first-author manuscript under review at the Journal of Artificial Intelligence Research.

RESEARCH INTERESTS

Reliable LLM systems: grounding answers in evidence, measuring where a model is wrong, and building checks that catch it. Evaluation methodology for machine learning: leak-free validation, small labelled sets and noisy labels. Multimodal learning across text, images and sensor data, with applications in health and education.

MANUSCRIPTS UNDER REVIEW

- **B. Roy** (first and corresponding author) and one co-author. Predicting Problematic Internet Use in Children: A Novel Methodology Leveraging Quadratic Weighted Kappa and Advanced Multi-Modal Feature Engineering. Under review, Journal of Artificial Intelligence Research (JAIR).
- Co-author. Real-Time Indian Sign Language Translation using Deep Learning. Under review, Pattern Recognition (Elsevier).

RESEARCH PROJECTS

Predicting problematic internet use in children | First-author manuscript, under review at JAIR; 2024 - 2025

- **Question:** can the severity of problematic internet use in children, an ordered 0 to 3 score built from a parent-reported questionnaire, be predicted from clinical, fitness, body-composition and sleep measurements and wrist-accelerometer recordings?
- **Data:** the Child Mind Institute's competition data: 3,960 participants, with wrist-accelerometer recordings for 996 of them, each recording reduced to 96 summary statistics; the classes are very uneven, with 34 severe cases among 2,736 labelled children.
- **Method:** treated the ordered score as regression and tuned three cut-points to maximise quadratic weighted kappa on out-of-fold predictions; compared voting ensembles of LightGBM, XGBoost and CatBoost under stratified cross-validation, keeping the questionnaire items that define the label out of the features.
- **Writing:** first and corresponding author of the 20-page manuscript, with one co-author.

Knee MRI abnormality detection with LLM-read report labels | RSNA challenge on Kaggle, 2026

- **Problem:** predict 12 knee abnormalities from multi-sequence MRI when only 58 of 4,407 training studies carry radiologist labels and the rest have only a free-text report, written in one of nine languages.
- **Labels:** used an LLM to read 4,276 reports against the organisers' annotation criteria and label 4,349 studies, after a pilot on the 58 radiologist-labelled reports had to beat a multilingual rules baseline: 0.90 against 0.78 macro ROC-AUC.
- **Data pipeline:** 617,004 DICOM slices ordered, windowed and cached at 336 px in a resumable build; up to six MRI sequences per study recognised from the series description, and right knees mirrored so that every study shares one frame.
- **Model:** a DINOv2 vision transformer fine-tuned on 2.5D slices in PyTorch, with 12 learned label queries attending across all slices of a study: 0.91 macro ROC-AUC against the report labels on a held-out fold of 1,102 studies, up from 0.885 with the smaller backbone.
- **Validation:** four folds grouped by report text so that duplicate reports never cross folds, with models scored against two references, the report labels and the radiologist labels.
- **Result:** a pseudo-label round lifted the four-fold models from 0.84 to 0.89 macro ROC-AUC on the 58 radiologist-labelled studies, scored out-of-fold; a second LLM re-read of the reports brought no gain and was dropped.
- **Compute:** two Kaggle T4 GPUs in resumable 8.5-hour sessions, with a pass bar written down before each experiment.

Mudra: Indian Sign Language recognition | Smart India Hackathon 2024, team project; basis of the co-authored manuscript

- **System:** turns Indian Sign Language signs captured on camera into text, built by the team with TensorFlow and MediaPipe for the Indian Sign Language problem statement of Smart India Hackathon 2024.
- **Data:** assembled the training set from two public Indian Sign Language datasets: 43,781 labelled images in 149 classes (26 letters, 9 digits and 114 words) and 18,863 sentence-level frames covering 97 sentences, with fixed train and validation split files so that every run used the same images.
- **Open problem:** the set is strongly imbalanced, with about 1,200 images for each letter or digit and a median of 6 for each word sign, so word-level recognition is a few-shot task.

RESEARCH AND INDUSTRY EXPERIENCE

Applied AI Intern | Interview Kickstart, Remote Apr 2026 - Oct 2026

Full end-to-end ownership of Feedback Loop, a live LLM platform that scores 3,000+ classes and drafts instructor feedback; built for about 100 programme managers, in pilot since July 2026

- **Designed and built a multi-stage LLM pipeline** (a whole-session map, 30-minute windows, synthesis and an adversarial self-check) that turns a 4-hour class recording into evidence-backed instructor feedback in about 10 minutes for about \$1.
- **Made every finding checkable:** each carries a transcript quote and a timestamp; for serious findings, code looks the quote up in the transcript and a second LLM pass tries to refute the finding; a programme manager approves every note before it is used.
- **Tested the vision check against a labelled answer key** (is the instructor's camera on; 136 screenshots from one class): accuracy was 84% with 20 frames per call and 99% with one frame per call, so each frame now gets its own call.
- **Audited the pipeline's repeatability:** six classes analysed twice agreed on the re-teach decision only once; traced it to a summary stage that read only 27 to 65% of long classes, raised its coverage to 100% on ten recordings and added 34 behavioural tests.
- **Compared 28 scoring formulas out of time:** fitted on five months (1,752 classes) and scored on the next three (1,027 classes) by AUC for predicting a low next class, with cluster-bootstrap standard errors; no alternative beat the active formula by more than the noise.
- **Measured how fragile a verdict is:** in a perturbation test on 2,784 classes, changing one learner's vote flipped the verdict of 31% of classes; a minimum-response rule cut that to 13%.
- **Estimated how far to trust small samples:** a low average becomes believable from about 5 votes; fitted the shrinkage strength (about 3 prior raters) and a beta prior for the yes/no vote on 8 months of classes.
- **Measured and cut cost:** traced a 4x rise in output tokens to the new model's built-in reasoning, then cached the 9.4K-token prompt block shared by every 30-minute window, cutting LLM cost 10-15% on long classes.
- **Built Ask AI, question answering over one class:** the whole transcript is attached and cached, each quote in an answer comes from the model's citation feature and its timestamp is looked up in code; a question costs about 15 cents.
- **One versioned SQL scoring function** with preview, activate and rollback, kept identical across SQL, Python and TypeScript by 120 shared test cases.
- **Built the data pipeline:** class ratings sync from Google Sheets to PostgreSQL three times a day; fingerprinting rows so that only changes are written made a sync about 25x faster, from 9 minutes to about 20 seconds.
- **Fixed a defect that mislabelled 67% of class records** (1,865 of the 2,780 then loaded) and parsed free-text cohort labels into 238 cohorts at 99.8% coverage.
- **Owned the platform end to end:** a Next.js app, a Dockerised FastAPI worker and PostgreSQL with row-level security, covered by 850+ automated tests; it scores 3,000+ classes across 8 courses three times a day.
- **Wrote the design, then took the evidence to the team:** a plain-English build plan fixing what is automatic and what a person decides, and reports for the manager and VP on why the formula was kept.
- **Built and deployed agent and RAG projects on Microsoft Foundry** for the company's Azure AI Engineer programme.

AI Engineer Intern | CloudCredits, Remote Jul 2025 - Oct 2025

Risk modelling on the company's own transaction and customer data

- **Deployed XGBoost and Random Forest risk models** on 5M+ transactions: 34% fewer false positives than the earlier model.
- **Engineered 12 risk indicators from 3.2M customer records**, lifting credit-scoring precision 27% and cutting default-prediction error 18%.

Internship project: implementation of a chatbot using natural language processing

- **Built a chatbot that answers questions about AI and data science by intent classification:** 29 intents with 257 example questions and 283 responses, preprocessed with NLTK tokenising and lemmatising and a stop-word list that keeps question words.
- **Trained a stacked bidirectional LSTM in Keras** (a 128-dimension embedding, two bidirectional LSTM layers and one LSTM layer, with batch normalisation, dropout and L2 regularisation), using class weights, early stopping and learning-rate reduction on plateau.
- **Expanded the 257 questions to 1,281 training samples** with rule-based rephrasing; the model reached 96.5% validation accuracy on a stratified 20% hold-out of the expanded set (256 samples).
- **Served it as a Streamlit chat app** that replies only when the model's confidence passes a threshold and otherwise asks the user to rephrase.

SELECTED PROJECTS

Kakehashi | FAR AWAY 2026 hackathon finalist, rank 14; solo github.com/roybishal362/Kakehashi

- **Built a fixed pipeline of LLM agents in plain Python** that turns a worker's profile or resume into a migration plan with source links: an LLM router picks the visa route, and five agents cover eligibility, jobs, application steps, a study plan and a cost summary.
- **Grounded the eligibility agent and the follow-up chat in BM25 retrieval** over a curated knowledge base of 26 facts, each with a source URL, and tested factual consistency: an LLM judge flagged 0 contradictions in 132 fact checks (22 gold facts, 3 personas, 2 runs each).
- **Served it as a FastAPI service with a Next.js front end:** each agent step streams to the browser as it runs, the job-search tool falls back to a clearly labelled cached sample when its API is unavailable, and 22 offline tests run in GitHub Actions on every push.

C-TRUST | Novartis NEST 2.0 National Semifinalist github.com/roybishal362/NEST_2.0_Hackthon

- **Rule-based specialist agents score each clinical trial's data-quality risk from 0 to 100:** each returns a risk level and a confidence, a confidence-weighted consensus counts safety findings three times as much, and a separate check flags inconsistencies across agents.
- **Engineering:** a FastAPI and pandas back end with a React and TypeScript dashboard; 550+ passing tests, 130+ of them property-based (Hypothesis).

VayuNetra | Air-quality forecasting for six Indian cities; independent project, 2026

- **Forecast PM2.5 up to 72 hours ahead for six Indian cities** with one gradient-boosted model per city, scored against a same-hour persistence baseline on a held-out final week: clear gains in four cities, little or none in coastal Chennai and Mumbai.
- **Built the platform around the forecast:** hourly modelled pollution and weather data for 59 localities, 36 features per forecast, an 18-endpoint FastAPI service with cached responses and a Next.js map dashboard.
- **Dropped a feature set that leaked the target:** target-hour chemistry made the measured gain jump, so it was left out of the shipped model.

Smart Product Pricing | Amazon ML Challenge 2025, top 8%, team github.com/roybishal362/Amazon_ML_2025

- **Designed the team's multimodal pipeline to predict the price of 75K products from text and images alone:** TF-IDF text features reduced with SVD, EfficientNetBo image embeddings (TensorFlow), brand target encoding and a LightGBM + XGBoost ensemble with pseudo-labelling.

Cricket scouting system | Rajasthan Royals SuperRR Selector Hackathon 2025, 4th place nationally

- **Analysed 602,992 ball-by-ball deliveries across 2,714 matches and 5,468 players** and engineered batting and bowling features to rank the 3,804 uncapped Indian players for scouting; solo entry.

LLM and retrieval projects | 2025

- **Customer support copilot:** one LLM call labels each ticket by topic, sentiment and priority, and a retrieval pipeline (LangChain, MiniLM embeddings, FAISS) answers 5 of 9 topics from crawled product documentation and lists the source pages.
- **Assessment recommender:** semantic search with all-MiniLM-L6-v2 embeddings in an exact FAISS index, with a custom evaluation script; MAP@3 0.73 and Precision@3 0.78 on 6 labelled queries over a catalogue of 37 assessments.

Other Kaggle competitions | 2024 - 2025

- **UM Game-Playing Strength of MCTS Variants (2024):** predicted match outcomes between Monte Carlo Tree Search agents from 233,234 game records with a LightGBM and CatBoost stack under cross-validation grouped by game ruleset; adapted from a public baseline.
- **RSNA Intracranial Aneurysm Detection (2025):** a PyTorch baseline that turns each DICOM series into a three-slice image and trains an EfficientNet-B3 with separate outputs for aneurysm presence and 13 artery locations.

HONOURS AND AWARDS

- **Smart India Hackathon 2024:** Runner-up at the Grand Finale with Mudra (Indian Sign Language problem statement; one of 5 teams selected nationally for it).
- **Rajasthan Royals SuperRR Selector Hackathon 2025:** 4th place nationally as a solo entrant (top 8 at the finale; 7,500+ participants from 39 countries).
- **Amazon ML Challenge 2025:** top 8% on the final leaderboard (team); designed the team's multimodal pipeline.
- **Novartis NEST 2.0 (2025-26):** National Semifinalist with C-TRUST, a rule-based multi-agent data-quality system for clinical trials.
- **FAR AWAY 2026 hackathon:** Finalist, rank 14, with Kakehashi (solo).
- **Kaggle:** 5 competitions across health data, medical imaging, game AI and LLM optimisation.
- **Coding practice:** 100+ problems solved on LeetCode; gold badge in SQL (5 stars) on HackerRank.

LEADERSHIP AND TEACHING

- **Google Developer Groups on Campus, Dr. D. Y. Patil Institute of Technology:** AI/ML Co-Lead; ran 8 workshops on generative AI, LLMs and retrieval-augmented generation for 400+ developers.

CERTIFICATIONS

- **Microsoft Certified:** Azure AI Apps and Agents Developer Associate (AI-103); Azure AI Fundamentals (AI-901).

TECHNICAL SKILLS

Programming: Python, SQL, TypeScript

Machine learning: PyTorch, TensorFlow and Keras, scikit-learn, XGBoost, LightGBM, CatBoost, vision transformers (DINOv2), EfficientNet, LSTM

LLM systems: multi-stage LLM pipelines, RAG (BM25, FAISS), AI agents, prompt caching, LLM evaluation, LangChain, Microsoft Foundry

Evaluation and statistics: held-out and grouped cross-validation, ROC-AUC, quadratic weighted kappa, bootstrap error, shrinkage priors, perturbation tests

Data and engineering: pandas, NumPy, NLTK, PostgreSQL, FastAPI, Next.js, Streamlit, Docker, pytest, Git, AI coding assistants (Claude Code)

LANGUAGES

English, Hindi, Bengali, Marathi